Assignment – Supply Chain Game

Assignment due on March 2nd at 10am

This assignment will recreate the decisions you should expect to make within a strategic supply chain management setting. It is a group activity underpinned by a competitive web-based simulation game that will run over three weeks.

The game will consist of three stages, each set at a different level of the platform.

Stage 1 – Trial Run (Week 3)

Monday 26th Jan (1pm) to Thursday 29th Jan (1pm)

This stage involves a non-assessed trial run, where you will have an opportunity to practice team coordination and get familiarised with the web-based platform. Your choices in this stage will not influence the marks that your team will receive.

Stage 2 – One Region Run (Week 4)

Monday 2nd Feb (1pm) to Sunday 7th Feb (1pm)

This stage will repeat the same scenario that we encountered in the trial run. However, this time, the simulation will last 6 days, and your performance will now be assessed.

Stage 3 – Network Run (Week 6)

Monday 16th Feb (1pm) to Sunday 21st Feb (1pm)

This scenario is set in the same world as the previous stages, but this time you will have the opportunity to control the facilities across all regions on the map.

Your assignment

All students should be assigned to a team. Throughout the assignment, you will have to co-ordinate with your team members, to select a production and supply strategy that will meet product demand. Ultimately, your objective is to maximize your overall profit by the end of the simulated period.

While all the teams will co-exist in the same world, your actions will not affect the outcomes of other player's strategies. You will still be able to compare your performance against others through an online leaderboard.

During the game, you will be able to control the following aspects of the simulation:

1. Increase your production output by establishing a new factory or by modifying your production capacity levels.
2. Construct new warehouses to increase stock levels and reduce the cost of long-distance deliveries to a region.

3. Modify the re-order point and shipment quantities or select a new transportation mode between factories and warehouses.

IMPORTANT: THERE IS NO UNDO BUTTON - your actions in the game are final. Some modifications will take some time to complete and incur costs, as in the real world.

Your Report

Each team will submit a report, approximately 10 pages long (excluding references and appendices). **The report is due March 2nd at 10am.**

In this report, you will need to present a log of the decisions that you made during the 2nd and 3rd stages of the game. Your discussion should detail the following:

- The **long-term decisions** that you made at the beginning of each stage (i.e., new factories, warehouses, capacity levels)
- A list of **short-term decisions** that you made throughout the simulation (i.e., reorder-points, shipping methods, network configuration).

You should accompany your discussion by an account of any models or formulas that you used, which guided your decision-making process.

Your report should conclude with a commentary of your overall performance and discuss any alternative choices that you could have made during the game to increase your performance.

Your Presentation

Each team will make a presentation on their strategy, performance, and lessons learnt throughout the exercise. Presentations will take place on **Feb 24th and Feb 26th** – these sessions should already be in your calendar.

Each team will have 10 minutes allocated, so 8-minutes presentation and 2 minutes Q&A. Presentation slides should be shared with Dr Jose Escribano by **5pm Feb 23rd.**

Mark-scheme

Your individual marks for this assignment will be calculated as follows:

- 40% of your mark will depend on your team's total profit in Stages 2 and 3.
- 45% of your mark will be based on the quality of your group report.
- 10% of your mark is based on your group presentation.
- 5% of your mark will be linked to a peer assessment exercise, in which every student will assess the contribution of every member of their teams (more details upon submission).

Register

You can find access and signup details in the accompanying document in blackboard. You must first signup in the accompanying sign up excel spreadsheet and then create a team.

It is imperative that every team registers in the online platform **before the end of Friday, 23rd Jan**. Once the simulation begins, you will **not** be able to make any modifications. Any team that does not register before **Friday 23rd January** will **NOT** participate in the trial run.

Platform Overview

We have prepared a video introduction that discusses the game mechanics and provides suggestions on the parts of the game that you should pay particular attention to. The video is made available through blackboard.

Stage 1 of the game will be a trial run that will help you understand how the game works. You can use your experience from Stage 1 to plan the remainder of the game.

Frequently Asked Questions

Do we need to refer to any external materials?

No. While using outside sources to further your understanding of the theory is always encouraged, the materials that we covered during weeks 1, 2 & 3 should provide all the tools you need to succeed in the game.

Something went wrong - I do not know if I submitted my changes correctly.

The simulation will ask for confirmation for most actions. You can obtain a list of previous actions that you carried out through the "History" tab on the main page of the simulation.

We made a mistake! Can we please restart the simulation?

No.

Hints and tips

- Note that you have limited resources in the simulation, and multiple uncoordinated changes can quickly diminish your funds!
- Do not press the refresh button immediately after changing a value, as it may cause the server to process your request twice (i.e., create another factory). Try to make one change at a time.
- It might be worth agreeing in advance within your group who will execute any action that was agreed by the team.
- Please collaborate with the rest of your team and develop a preliminary plan before the simulation starts.
- Make sure that you study the data that has been provided!
- Do take care to monitor the progress of the simulation consistently.
- You have 7-8 people in your group. Share the workload and plan ahead.
- Do note that another source of income corresponds to the **interest earned from accumulated cash.**

Good luck!